Naive Bayes Classifier

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Recall:

Bayes' Rule:

$$P(A|B) = \frac{P(A)\,P(B|A)}{P(B)}$$

$P(A|B)$ is the posterior probability of A given B.
$P(B|A)$ is the likelihood of A given B.
$P(A)$ and $P(B)$ are the prior probabilities.

Consider the following binary classification data:

X_1	X_2	Y
0	2	0
0	1	1
1	0	1
0	0	1
2	2	0
1	1	0
0	2	1
2	0	0
0	2	0
1	0	0

Here $X_1, X_2 \in \{0,1,2\}$ and $Y \in \{0,1\}$.

Q: What will be the prediction for a new sample with $X_1 = 0, X_2 = 2$?

What will be the values of $P(Y = 0 \mid X_1 = 0, X_2 = 2)$ and $P(Y = 1 \mid X_1 = 0, X_2 = 2)$?

$$P(Y = 0 \mid X_1 = 0, X_2 = 2) = \alpha\, P(Y = 0)\, P(X_1 = 0, X_2 = 2 \mid Y = 0)$$
$$= \alpha\, \frac{6}{10} \cdot \frac{2}{6}$$
$$= \alpha\, (0.2).$$

$$P(Y = 1 \mid X_1 = 0, X_2 = 2) = \alpha\, P(Y = 0)\, P(X_1 = 0, X_2 = 2 \mid Y = 0)$$
$$= \alpha\, \frac{4}{10} \cdot \frac{1}{4}$$
$$= \alpha\, (0.1).$$

Hence the prediction for a new sample with $X_1 = 0, X_2 = 2$ will be $Y = 0$.

The issue with the above way of computation:

Suppose there are n features, $X_1, X_2, \ldots, X_n$ which can take $r_1, r_2, \ldots, r_n$ possible values respectively. Then the number of possible combinations of feature values is $r_1 \cdot r_2 \cdot \cdots \cdot r_n$.

For example, if each X_i $(1 \leq i \leq n)$ is a Boolean random variable, then the number of possible combinations of feature values is 2^n.

Thus, computing $P(X_1, X_2, \ldots, X_n \mid Y)$ is usually impossible for large feature spaces.

To get around this issue, a **naive** assumption of ***Conditional Independence*** is used.

Given a (binary or multiclass) classification problem with input variables (features) $X_1, X_2, \ldots, X_n$, and target variable Y, we have

$$P(Y, X_1, X_2, \ldots, X_n) = P(Y) \cdot P(X_1, X_2, \ldots, X_n \mid Y).$$

The **assumption** of conditional independence:

$$P(X_1, X_2, \ldots, X_n \mid Y) = P(X_1 \mid Y) \cdot P(X_2 \mid Y) \cdot \cdots \cdot P(X_n \mid Y)$$

$$= \prod_{i=1}^{n} P(X_i \mid Y).$$

Hence,

$$P(Y, X_1, X_2, \dots, X_n) = P(Y) \cdot \prod_{i=1}^{n} P(X_i \mid Y).$$

We have

$$P(Y \mid X_1, X_2, \dots, X_n) = \frac{P(Y, X_1, X_2, \dots, X_n)}{P(X_1, X_2, \dots, X_n)}$$

$$= \alpha\, P(Y, X_1, X_2, \dots, X_n)$$

$$= \alpha\, P(Y) \cdot \prod_{i=1}^{n} P(X_i \mid Y).$$

(by assuming conditional independence)

Example: Text Classification

Consider the following dataset of news articles labeled into different categories:

Sl. No.	News Headline	Category
1	Team wins final	Sports
2	Star player scores twice	Sports
3	Company signs major deal	Business
4	Market sees big investment	Business
5	Storm expected tomorrow	Weather
6	Weather department issues alert	Weather

Classify the news headline “Star signs major deal” using Naive Bayes Classifier.

Set of all words (Vocabulary) from the dataset:

V = {*team, wins, final, star, player, scores, twice, company, signs, major, deal, market, sees, big, investment, storm, expected, tomorrow, weather, department, issues, alert***}**

|V| = 22.

Let Y denote the random variable corresponding to the news category. For a text/document $\boldsymbol{x}$, we have

$$P(Y \mid \boldsymbol{x}) = \frac{P(Y)P(\boldsymbol{x} \mid Y)}{P(\boldsymbol{x})}$$

$$= \alpha\, P(Y)P(\boldsymbol{x} \mid Y).$$

We will look at two types of Naive Bayes Classifier and how they compute the above probability for the text $\boldsymbol{x}$ =“Star signs major deal”.

The following are two types of Naive Bayes Classifiers:

1. Bernoulli Naive Bayes Classifier
2. Multinomial Naive Bayes Classifier

1. Bernoulli Naive Bayes Classifier

- Feature representation: Binary (0/1).
- It only looks at whether a feature (word) appears or not in the document.
- Useful for short texts or when word *occurrence* is more important than *frequency*. (Ex.: News headlines, E-mail subject lines, etc.)

Let X_i be the Boolean random variable representing the presence or absence of ith word in the text/document $\boldsymbol{x}$ (taking values $x_i = 1$ or $x_i = 0$ respectively). Note that this can be modelled using Bernoulli distribution. We have

$$P(\boldsymbol{x} \mid Y = c) = \prod_{\boldsymbol{i}} P(x_i = 1 \mid Y = c)^{x_i} \, P(x_i = 0 \mid Y = c)^{1-x_i}$$

For each i, $P(X_i \mid Y = c)$ is computed as follows:

$$P(x_i = 1 \mid Y = c) = \frac{\text{Number of documents in Category } c \text{ containing } i^{\text{th}} \text{ word}}{\text{Number of documents in Category } c}.$$

$$P(x_i = 0 \mid Y = c) = \frac{\text{Number of documents in Category } c \text{ not containing } i^{\text{th}} \text{ word}}{\text{Number of documents in Category } c}.$$

For the news headline “Star signs major deal”,

$$P(Y=Sports \mid x_i = 1 \text{ for } i \in \{4,9,10,11\}, x_i = 0 \text{ for } i \notin \{4,9,10,11\})$$

$$= \alpha\, P(Y=Sports) \prod_{i\in\{4,9,10,11\}} \mathrm{P}(x_i = 1 \mid Y=Sports) \prod_{i\notin\{4,9,10,11\}} \mathrm{P}(x_i = 0 \mid Y=Sports)$$

$$= \alpha\, \frac{2}{6}\left(\frac{1}{2}\cdot\frac{0}{2}\cdot\frac{0}{2}\cdot\frac{0}{2}\right)\left(\left(\frac{1}{2}\right)^{6}\cdot\left(\frac{2}{2}\right)^{12}\right)$$

$$= 0.$$

Note that even a single probability value being zero “wipes out” the entire probability computation.

2. Multinomial Naive Bayes Classifier

- Feature representation: Counts (0,1,2,…).
- It looks at how many times a feature (word) occurs in a document.
- Suitable for longer documents. (Ex.: News articles, Product reviews, E-mail content, etc.)

Let $\boldsymbol{x}$ be a document of length N.
For each i $(1 \leq i \leq |V|$, where V is the vocabulary):
let X_i be the categorical random variable representing the count of ith word in the text/document $\boldsymbol{x}$ (taking values $x_i = 0, 1, 2, \ldots$).
let $P(w_i \mid Y = c)$ denote the probability of occurrence of ith word in Category c documents.
We have $N = \sum_i x_i$, the total count of words in a text/document. Note that this can be modelled using the categorical distribution.
We have

$$P(\boldsymbol{x} \mid Y = c) = \frac{N!}{\prod_i x_i!} \prod_i P(w_i \mid Y = c)^{x_i} .$$

In practice, the multinomial coefficient $\frac{N!}{\prod_i x_i!}$ is often ignored for classification since it is the same across all classes for a fixed $\boldsymbol{x}$.

For each i, $P(w_i \mid Y = c)$ is computed as follows:

$$P(w_i \mid Y = c) = \frac{\text{Total count of } i^{\text{th}} \text{ word in Category } c}{\text{Total number of words in Category } c}.$$

For the news headline “Star signs major deal”,

$P(Y=Sports \mid x_i = 1 \text{ for } i \in \{4,9,10,11\}, x_i = 0 \text{ for } i \notin \{4,9,10,11\})$

$$= \alpha\, P(Y=Sports) \prod_{i \in \{4,9,10,11\}} P(w_i \mid Y=Sports)$$

$$= \alpha\, \frac{2}{6} \left(\frac{1}{7} \cdot 0 \cdot 0 \cdot 0 \right)$$

$$= 0.$$

Note that even a single probability value being zero “wipes out” the entire probability computation.

Laplace Smoothing:

Laplace smoothing is a technique used to handle zero probabilities in Naive Bayes classification.

It ensures that every word has a non-zero probability, even if it hasn't appeared in the training data for a given class.

After Laplace Smoothing in Bernoulli Naive Bayes, $P(X_i \mid Y = c)$ is computed for each i as follows:

$$P(x_i = 1 \mid Y = c) = \frac{(\text{Number of documents in Category } c \text{ containing } i^{\text{th}} \text{ word}) + 1}{(\text{Number of documents in Category } c) + 2}$$

$$P(x_i = 0 \mid Y = c) = \frac{(\text{Number of documents in Category } c \text{ not containing } i^{\text{th}} \text{ word}) + 1}{(\text{Number of documents in Category } c) + 2}$$

After Laplace Smoothing in Multinomial Naive Bayes, $P(w_i \mid Y = c)$ is computed for each i as follows:

$$P(w_i \mid Y = c) = \frac{(\text{Total count of } i^{\text{th}} \text{ word in Category } c) + 1}{(\text{Total number of words in Category } c) + |\mathbf{V}|},$$

where $|\mathbf{V}|$ is the total number of words in the vocabulary.

Let us now compute the required probabilities in both Bernoulli Naive Bayes and Multinomial Naive Bayes by applying Laplace Smoothing.

Bernoulli Naive Bayes:

For the news headline “Star signs major deal”,

$$P(Y=Sports \mid x_i = 1 \text{ for } i \in \{4,9,10,11\}, x_i = 0 \text{ for } i \notin \{4,9,10,11\})$$

$$= \alpha\, P(Y=Sports) \prod_{i\in\{4,9,10,11\}} \mathrm{P}(x_i = 1 \mid Y=Sports) \prod_{i\notin\{4,9,10,11\}} \mathrm{P}(x_i = 0 \mid Y=Sports)$$

$$= \alpha\, \frac{2}{6}\left(\frac{(1+1)}{(2+2)}\cdot\frac{(0+1)}{(2+2)}\cdot\frac{(0+1)}{(2+2)}\cdot\frac{(0+1)}{(2+2)}\right)\left(\left(\frac{1+1}{2+2}\right)^{6}\cdot\left(\frac{2+1}{2+2}\right)^{12}\right)$$

$$\approx \alpha\,(1.2889\times10^{-6}).$$

Similar computations for $P(Y = Business \mid \boldsymbol{x})$ and $P(Y = Weather \mid \boldsymbol{x})$ give:
$P(Y = Business \mid \boldsymbol{x}) \approx \alpha\ (7.7335\times10^{-6})$
$P(Y = Weather \mid \boldsymbol{x}) \approx \alpha\ (4.2964\times10^{-7})$.

Hence, $P(Y \mid \boldsymbol{x}) = \alpha \left\langle 1.2889\times10^{-6},\ 7.7335\times10^{-6},\ 4.2964\times10^{-7} \right\rangle$.

After normalizing, we get the following probability distribution:
$P(Y \mid \boldsymbol{x}) \approx \langle 0.1364,\ 0.8182,\ 0.0454 \rangle$.

Hence the predicted category is ***Business***.

Multinomial Naive Bayes:

For the news headline “Star signs major deal”,

$P(Y=Sports \mid x_i = 1 \text{ for } i \in \{4,9,10,11\}, x_i = 0 \text{ for } i \notin \{4,9,10,11\})$

$$= \alpha\, P(Y=Sports) \prod_{i \in \{4,9,10,11\}} \mathrm{P}(w_i \mid Y=Sports)$$

$$= \alpha\, \frac{2}{6} \left(\frac{(1+1)}{(7+22)} \cdot \frac{(0+1)}{(7+22)} \cdot \frac{(0+1)}{(7+22)} \cdot \frac{(0+1)}{(7+22)} \right)$$

$$\approx \alpha\, (9.4258 \times 10^{-7}).$$

Similar computations for $P(Y = \mathit{Business} \mid \boldsymbol{x})$ and $P(Y = \mathit{Weather} \mid \boldsymbol{x})$ give:
$P(Y = \mathit{Business} \mid \boldsymbol{x}) \approx \alpha\ (3.2922 \times 10^{-6})$
$P(Y = \mathit{Weather} \mid \boldsymbol{x}) \approx \alpha\ (4.7129 \times 10^{-7})$.

Hence, $P(Y \mid \boldsymbol{x}) = \alpha \left\langle 9.4258 \times 10^{-7},\ 3.2922 \times 10^{-6},\ 4.7129 \times 10^{-7} \right\rangle$.

After normalizing, we get the following probability distribution:
$P(Y \mid \boldsymbol{x}) \approx \langle 0.2003,\ 0.6996, 0.1001 \rangle$.

Doing the above computations in log space is advantageous, because multiplying a large number of small values can lead to significant rounding error.

Applying a log transform converts the product into a sum, and reduces the effect of this rounding error.

$$\log_e\left(P(Y=\text{Sports} \mid x)\right) = \log_e \alpha + \log_e P(Y=\text{Sports}) + \sum_i x_i \log_e P(w_i \mid Y=\text{Sports}).$$

$$= \log_e \alpha + \log_e \frac{2}{6} + \left(\log_e \frac{2}{29} + \log_e \frac{1}{29} + \log_e \frac{1}{29} + \log_e \frac{1}{29}\right)$$

$$= \log_e \alpha - 13.8746.$$

$$P(Y=\text{Sports} \mid x) = e^{\log_e \alpha} \cdot e^{-13.8746}.$$

$$\approx \alpha \left(9.4262 \times 10^{-7}\right).$$

Similarly, we can compute

$$P(Y = \text{Business} \mid x) \text{ and } P(Y = \text{Weather} \mid x)$$

and normalize the values to get the probability distribution $P(Y \mid x)$.